

1. AV roundabout

WHAT YOU SEE

A green highway 'AV' sign at the center of a traffic roundabout, with cars circling endlessly — the roundabout IS the re-entry loop spinning inside the AV node.

WHAT IT ENCODES

AVNRT (atrioventricular nodal re-entrant tachycardia) is the most common paroxysmal SVT (~60%), with the re-entry circuit contained entirely within/around the AV node using two functionally distinct pathways: a slow pathway (short refractory period, slow conduction) and a fast pathway (long refractory period, fast conduction). The roundabout = electricity circling the AV node. Typical rate is 150-250 bpm with a narrow regular QRS.

BOARD TRAP

Confusing AVNRT with AVRT is the trap — AVRT (e.g., WPW) uses an accessory pathway OUTSIDE the node creating a macro-re-entrant circuit, whereas AVNRT's circuit is micro-re-entrant within the node itself; this distinction changes the ablation target and the danger of pre-excited AF.

2. Slow down, fast up

WHAT YOU SEE

Black road banners reading 'SLOW DOWN, FAST UP' arcing over the circle — the impulse goes DOWN the slow pathway and back UP the fast pathway (typical slow-fast AVNRT).

WHAT IT ENCODES

Typical ('slow-fast') AVNRT, ~90% of cases, conducts ANTEGRADE down the slow pathway and RETROGRADE up the fast pathway. Because retrograde fast conduction reaches the atria almost simultaneously with ventricular activation, the P wave is buried in or just after the QRS (very short RP interval). Atypical ('fast-slow') AVNRT reverses this with a long RP.

BOARD TRAP

Assuming every AVNRT is slow-fast is the trap — atypical fast-slow AVNRT produces a long-RP tachycardia that can be mistaken for atrial tachycardia or AVRT; the buried/short-RP P wave is the marker of the common typical form.

3. Slow pathway

WHAT YOU SEE

A yellow 'SLOW PATH' road sign marking the slow lane feeding the roundabout — the slow-conducting input that is the safe ablation target.

WHAT IT ENCODES

The slow pathway conducts antegrade in typical AVNRT and sits near the tricuspid annulus/coronary sinus os (the inferior 'slow' input to the node). It is the ablation target precisely because eliminating it breaks the circuit while preserving normal fast-pathway AV conduction, giving ~95% cure with low complication risk.

BOARD TRAP

Ablating the FAST pathway is the trap — fast-pathway ablation carries a much higher risk of complete heart block (it lies near the compact node/His), so the slow pathway is deliberately targeted instead.

4. Premature beat trigger

WHAT YOU SEE

PREMATURE car at barrier, FAST blocked

WHAT IT ENCODES

A premature atrial complex (PAC) initiates AVNRT: arriving when the fast pathway is still refractory (its long refractory period), the impulse is blocked there and forced down the slow pathway; by the time it reaches the bottom, the fast pathway has recovered and conducts retrograde, closing the re-entry loop. This is why AVNRT starts and stops abruptly ('paroxysmal').

BOARD TRAP

Thinking AVNRT needs structural heart disease to start is the trap — it classically occurs in young, otherwise healthy patients (women > men), triggered by a simple PAC, caffeine, or stress, not by ischemia or cardiomyopathy.

5. Pseudo-r' in V1

WHAT YOU SEE

An ECG billboard pointing to a small upward 'PSEUDO-r' bump at the end of the QRS in lead V1 — the retrograde P masquerading as an RBBB r-prime.

WHAT IT ENCODES

In typical AVNRT, the retrograde P wave is buried at the end of the QRS and deflects upward in V1, mimicking an incomplete RBBB 'r-prime' (pseudo-r'). Comparing the tachycardia ECG to a baseline sinus ECG reveals this 'extra' deflection is actually the retrograde P, a classic AVNRT clue.

BOARD TRAP

Reading the pseudo-r' as genuine RBBB is the trap — it disappears in sinus rhythm because it is a retrograde P, not a true conduction-system finding; missing this misclassifies the tachycardia.

6. Pseudo-S inferior

WHAT YOU SEE

PSEUDO-S II, III, aVF

WHAT IT ENCODES

The same retrograde atrial activation produces a pseudo-S wave (terminal negative deflection) in inferior leads II, III, and aVF, reflecting near-simultaneous atrial and ventricular activation in slow-fast AVNRT. Pseudo-r' in V1 plus pseudo-S inferiorly together strongly support typical AVNRT.

BOARD TRAP

Mistaking the pseudo-S for a true inferior conduction abnormality is the trap — like the pseudo-r', it is a retrograde P that vanishes in sinus rhythm; it indicates a short RP interval typical of AVNRT, not inferior pathology.

7. Cannon A waves

WHAT YOU SEE

CANNON A label, distressed driver

WHAT IT ENCODES

Because the atria and ventricles contract almost simultaneously in typical AVNRT, the right atrium contracts against a closed tricuspid valve, producing regular cannon A waves visible in the jugular venous pulse (the 'frog sign'). Patients often report rapid regular neck pounding/palpitations.

BOARD TRAP

Interpreting cannon A waves as a sign of complete AV dissociation/3rd-degree block is the trap — in AVNRT they are REGULAR (every beat) from synchronized AV contraction, whereas in AV dissociation/VT they are IRREGULAR/intermittent.

8. Adenosine terminates

WHAT YOU SEE

ADENOSINE listed on the crowned 'first-line' car — the rapid-push AV-nodal blocker that snaps the re-entry circuit.

WHAT IT ENCODES

AVNRT is AV-node-dependent, so AV-nodal blockade terminates it. After failed vagal maneuvers, give adenosine as a rapid IV push: 6 mg, then 12 mg, then 12 mg if needed, each followed by a saline flush. Adenosine transiently blocks the AV node (half-life <10 seconds), interrupting the circuit. Warn the patient about a brief sense of chest discomfort/impending doom/flushing.

BOARD TRAP

Giving adenosine for a wide-complex or irregular tachycardia without caution is the trap — in pre-excited AF (WPW), AV-nodal blockers including adenosine can accelerate conduction down the accessory pathway and precipitate VF; adenosine is for regular narrow-complex SVT.

9. Valsalva terminates

WHAT YOU SEE

VALSALVA on the maneuver list — bearing down raises vagal tone to brake the AV node and break the loop.

WHAT IT ENCODES

Vagal maneuvers are first-line for stable AVNRT: the Valsalva maneuver and carotid sinus massage increase vagal tone, slowing AV-node conduction enough to break the circuit. The modified Valsalva (strain then lie supine with passive leg raise) significantly improves conversion rates over the standard technique.

BOARD TRAP

Performing carotid sinus massage without checking for bruits/recent TIA-stroke is the trap — it risks dislodging plaque and causing a stroke in older patients with carotid disease; the modified Valsalva is the safer universal first move.

10. AV block terminates

WHAT YOU SEE

Officer flag AV BLOCK TERMINATES

WHAT IT ENCODES

Because the AV node is an obligatory part of the AVNRT re-entry circuit, anything that transiently blocks the node (vagal maneuvers, adenosine, IV beta-blockers, or IV non-dihydropyridine CCBs) terminates the arrhythmia. This AV-node dependence is the defining therapeutic vulnerability of AVNRT.

BOARD TRAP

Expecting AV-node blockade to terminate every SVT is the trap — atrial tachycardia and atrial flutter continue in the atria despite AV block (you simply see slower ventricular response unmasking flutter waves), whereas AVNRT/AVRT actually stop because the node is part of their circuit.

11. Ablation cure

WHAT YOU SEE

A gold STAR labelled 'ABLATION ~95%' — the gold star marks slow-pathway ablation as the definitive ~95% cure.

WHAT IT ENCODES

Slow-pathway catheter ablation is the definitive long-term cure for recurrent or drug-refractory AVNRT, with ~95-97% success and low recurrence. It is preferred for patients who want to avoid lifelong medication. The main procedural risk is inadvertent AV block (~1%) requiring a pacemaker, which is why the slow pathway (not the fast) is targeted.

BOARD TRAP

Reserving ablation only for failed drug therapy in every patient is the trap — for frequent or highly symptomatic AVNRT, ablation is a reasonable first-line definitive option given its high cure rate, not merely a last resort.

12. BB / diltiazem / flecainide

WHAT YOU SEE

Beta bugle, calci-yum diltiazem, flecainide bottle

WHAT IT ENCODES

For chronic suppression/prophylaxis of recurrent AVNRT in patients declining ablation: AV-nodal blockers first — beta-blockers or non-dihydropyridine calcium channel blockers (diltiazem, verapamil). Class IC agents (flecainide, propafenone) are second-line and only for structurally normal hearts (no CAD/LVH).

BOARD TRAP

Combining a beta-blocker WITH IV verapamil/diltiazem, or using non-DHP CCBs in reduced-EF heart failure, is the trap — the combination causes profound bradycardia/AV block and hypotension, and non-DHP CCBs are negative inotropes contraindicated in HFrEF.

13. Accelerated junctional, no tx

WHAT YOU SEE

ACCELERATED JUNCTIONAL 50-100, NO TX

WHAT IT ENCODES

Accelerated junctional rhythm (rate 60-100 bpm, faster than the normal junctional escape of 40-60) arises from enhanced automaticity of the AV junction, often from digoxin toxicity, post-cardiac surgery, ischemia, or myocarditis. It is usually benign — management is to treat the underlying cause (e.g., stop digoxin, correct electrolytes), often with no specific antiarrhythmic.

BOARD TRAP

Treating accelerated junctional rhythm like AVNRT with adenosine/ablation is the trap — it is an automatic (not re-entrant) rhythm, so AV-nodal blockade and cardioversion do not fix it; the answer is to find and treat the cause, classically digoxin toxicity.

14. JET — pediatric/post-op

WHAT YOU SEE

Infant in gown 'JET / POST-OP' near jet plane

WHAT IT ENCODES

Junctional ectopic tachycardia (JET) is an automatic junctional tachycardia of infants and children, classically appearing after congenital cardiac surgery. Unlike re-entrant AVNRT it does not respond to cardioversion or adenosine; management focuses on cooling/hypothermia, avoiding catecholamines, correcting electrolytes, and amiodarone if needed.

BOARD TRAP

Cardioverting or giving adenosine for JET is the trap — being an automatic focus, it ignores both; the post-cardiac-surgery infant with an incessant narrow-complex tachycardia needs supportive cooling and amiodarone, not shock.

15. Diuresis / ANP after SVT

WHAT YOU SEE

ANP / DIURESIS droplet ladder

WHAT IT ENCODES

Prolonged SVT raises atrial pressure and stretch, releasing atrial natriuretic peptide (ANP), which produces a characteristic post-tachycardia polyuria/diuresis as the rhythm terminates. This benign 'urinary frequency after palpitations stop' is a clinical clue that the episode was a supraventricular tachycardia.

BOARD TRAP

Attributing post-SVT polyuria to a primary renal or endocrine problem is the trap — it is simply ANP-mediated diuresis from atrial stretch during the tachycardia, a recognized harmless sign, not a sign of kidney disease.
